MIL-A-62488(AT)
28 August 1985
SUPERSEDING
(see 6.3)

MILITARY SPECIFICATION

AMPLIFIER, AUDIO FREQUENCY

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an audio frequency amplifier (hereafter referred to as a distribution unit) used on military ground vehicles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5820

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SPECIFICATIONS
FEDERAL

QQ-A-367 - Aluminum Alloy Forgings.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-4158 - Electronic Equipment, Ground; General Requirements for.
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-C-6021 - Castings, Classification and Inspection of.
MIL-H-6088 - Heat Treatment of Aluminum Alloys.
MIL-I-6869 - Impregnants for Aluminum Alloy and Magnesium Alloy Castings.
MIL-H-6875 - Heat Treatment of Steel, Process for.
MIL-F-7190 - Forgings, Steel, for Aircraft and Special Ordnance Applications.
MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, and Aluminum and Aluminum Alloys.
MIL-P-11268 - Parts, Materials, and Processes Used in Electronic Equipment.
MIL-M-13231 - Marking of Electronic Items.
MIL-F-18264 - Finishes: Organic, Weapons System, Application and Control of.
MIL-C-22750 - Coating, Epoxy Polyamide.
MIL-I-46058 - Insulating Compound, Electrical (for Coating Printed Circuit Assemblies).
MIL-P-53022 - Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.

STANDARDS
FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-130 - Identification of Marking of US Military Property.
MIL-STD-252 - Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices.

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MIL-STD-276	- Impregnation of Porous Non-Ferrous Metal Castings.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-642	- Identification Marking.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1275	- Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.
MIL-STD-45662	- Calibration Systems Requirements.

DEPARTMENT OF DEFENSE

DOD-STD-100	- Engineering Drawing Practices.

HANDBOOKS
MILITARY

MIL-HDBK-5	- Metallic Materials and Elements for Aerospace Vehicle Structures.
MIL-HDBK-275	- Guide for Selection of Lubricant Fluids and Compounds for use in Flight Vehicles and Components.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

12310227	- Distribution Unit Assembly.
12310238	- Plate, Identification.

ELECTRONICS AND COMMUNICATIONS COMMAND

SC-D-48664	- Bracket Assembly.

PUBLICATIONS
DEPARTMENT OF DEFENSE

DOD 4120.3-M	- Defense Standardization and Specification Program Policies, Procedures and Instructions.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish distribution units which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent distribution units delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. The distribution unit materials shall comply with applicable portions of MIL-E-4158 or MIL-P-11268 as specified herein. Materials shall conform to MIL-HDBK-5 for physical properties when the materials are covered. If selected materials or combinations of materials are not included in MIL-HDBK-5, subcontractor accepted physical, chemical, and mechanical properties of the materials shall be documented. The use of strategic or critical materials shall be avoided.

3.2.1 Fungus proof materials. To the greatest extent practical, the materials used in the distribution unit shall be non-nutrients for fungi in accordance with MIL-STD-454, requirement 4. If materials that are nutrients for fungi must be utilized, such materials shall be approved by the Government (see 4.6.1).

3.2.2 Potting compounds. Potting compounds employed in the distribution unit shall conform to MIL-STD-454, requirement 47 (see 4.6.1).

3.2.3 Insulating compounds. Electrical coating for printed circuit assemblies shall conform to MIL-I-46058 (type ER, grade S preferred) (see 4.6.1).

3.2.4 Lubricants. Lubricants and lubrication practices shall conform to MIL-STD-454, requirement 43. MIL-HDBK-275 shall be used as a guide in the selection of lubricants. Lubricants shall function satisfactorily throughout the temperature range of their environment. Choice of lubricants shall:

a. Be compatible with non-metallic components.
b. Not damage finishes adjacent to location of lubricant applications.
c. Eliminate the need for frequent lubrication.

Oils or grease shall not be permitted to deteriorate paint systems. Where required, designs shall specify, or provide the absence of a paint system, in areas where oil and grease may build up. Dry lubrication shall be used on aluminum only after anodizing (see 4.6.1).

3.2.5 Metal parts. All ferrous metal parts shall be of corrosion resistant type or treated in a manner to render them resistant to corrosion in accordance with MIL-STD-454, requirement 15. Suitable protection against atmospheric corrosion and dissimilar metals and galvanic corrosion, as defined in MIL-STD-889, shall be provided. Magnesium shall not be used

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unless specifically authorized. Structural materials shall be selected after considering resistance to fatigue, susceptibility to stress corrosion damage, notch sensitivity, fracture toughness, fabrication methods, and availability (see 4.6.1).

3.2.6 Castings. Castings used in the distribution unit shall conform to MIL-C-6021. All aluminum castings subjected to differential pressure applications whereby leakage may occur shall be impregnated in accordance with MIL-STD-276, using type II impregnating material in accordance with MIL-I-6869. Castings which have been impregnated shall be marked "IMP". The impregnation shall be to correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity (see 4.6.1).

3.2.7 Non-metallic parts. Non-metallic parts shall be designed to minimize deterioration caused by exposure to sunlight, sand particles, microorganisms, moisture, temperature extremes, fuel, hydraulic and lubricating oil, grease, and salt degradation association with abrasion, load, or exposure-induced deterioration that can endanger or jeopardize the function of the system (see 4.6.1).

3.3 Design and construction. The distribution unit shall be fabricated and assembled in accordance with Drawing 12310227 and the drawings, parts list, and other documents listed thereon. The distribution unit shall meet the requirements of MIL-P-11268 or paragraphs 3.2 through 3.7 of MIL-E-4158 unless otherwise specified herein (see 4.5.2, 4.6.1 and 4.6.2).

3.3.1 Preconditioning. Each unit shall be preconditioned to ensure proper workmanship and to exclude infant mortality failures. The unit shall be continuously operated for a period of 48 hours with a 82 ohm load at pin C of J504 and 0.22 V rms, 1000 Hz at pin K of J501 at a temperature of 95°F to 113°F (40 $\pm$ 5°C) and with an input voltage of 25.5 V dc. While the unit is in the operating mode, vibration shall be performed as follows: 2.2 g $\pm$ 10% at any non-resonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. This vibration shall be applied for 10 minutes during each hour of equipment operating time. After preconditioning, there shall be no repair, cleaning, realignment, or other changes prior to acceptance inspection. In the event of failure to comply with performance requirements, corrective action shall be taken and the test segment during which failure occurred shall be repeated.

3.3.2 Size and weight. The distribution unit shall be 9.75 inches (in) [24.8 centimeters (cm)] long by 5.75 in (14.6 cm) wide by 3.75 in (9.5 cm) deep. The weight of the distribution unit shall not exceed 8 lb [3.6 kilograms (kg)] (see 4.6.3).

3.3.3 Controls and interfaces.

3.3.3.1 Interface definition. The distribution unit shall interface with a variety of audio accessories, radio sets (receive-transmit), receivers (radio receive only), and field wire line. Primary interface details shall be as specified in figure 1. When radio sets are used with the intercom, they are connected to the distribution unit thru J501, J503, J509 and J511 receptacles. All four radio sets may be of the receive-transmit type. The

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retransmission function requires that two radio sets be connected to receptacles J501 and J503. This function serves as a radio relay to increase the range between these two external radio sets. When radio sets (receive-transmit) are not used, direct current (dc) power for the intercom is supplied through receptacle J508. The radio output of an auxiliary receiver may be connected to the audio input binding posts to enable monitoring. This monitoring cannot be disabled by intercom controls. Field wire lines may be connected from a local battery telephone to the line binding posts to enable two-way communication between any crew member within reach of the distribution unit's LINE/PRESS TO TALK pushbutton switch. Monitoring field wire line communication is automatic and cannot be disabled by intercom controls (see 4.6.3).

3.3.3.2 On-off switch. A circuit breaker switch shall be provided to control power to the intercom system. This switch shall control power from all sources (see 4.6.3).

3.3.3.3 Line jacks. A pair of binding posts shall be provided to permit connection of a standard field telephone line (see 4.6.3).

3.3.3.4 Auxiliary input. A pair of binding posts shall be provided as an auxiliary input to the intercom channel (see 4.6.3).

3.3.3.5 Intercom cable assembly. A "standard" special purpose cable assembly (CX-4723/VRC) of the proper length shall be used to interconnect the distribution unit and crew control units. The cables shall be terminated by an eighteen-terminal male connector at both ends. Pin connections for this cable shall be as specified in figure 2 (see 4.6.3).

3.3.3.6 Power indicator. A green indicator lamp shall be provided to indicate when power is on. When the lens cap is rotated one quarter turn counterclockwise, the light shall not shine through the lens (see 4.6.3).

3.3.3.7 Intercom accentuated switch. When in the ON position, this switch shall reduce the audio level from all radio receivers (see 4.6.3).

3.3.3.8 Radio transmission selector switch. This switch shall control the transmitter keying lines of the radios as follows (see 4.6.3):

a. Commander and Crew (Position 1). Any crew control unit may initiate transmissions on any radio. If two or more crew members select a single radio, all may transmit on that radio.

b. Commander Only (Position 2). Only the crew control unit or units connected to J504 of the distribution unit may initiate radio transmissions.

c. Listening Silence (Position 3). No crew control unit shall be permitted to initiate radio transmissions.

3.3.3.9 Radio retransmission switch (see 4.6.3).

a. In NORMAL position, all radios operate independently.

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b. In RETRANSMISSION position, receive audio, transmit audio, and keying lines of the radios connected to J501 and J503 of the distribution unit shall be interconnected such that signals received by one radio are automatically retransmitted by the other radio. The connections shall be configured such that any crew control unit may monitor the audio passing between the radios, but no crew control unit may initiate transmission on either radio.

3.3.3.10 Line switch. Audio from a wire telephone line connected to the line terminals shall be monitored continuously by the intercom channel when the line switch is in the LISTEN position. When the switch is placed in the TALK position, audio from the intercom channel shall be transmitted to the telephone line. The switch shall be a spring loaded, momentary type, biased to the LISTEN position (see 4.6.3).

3.3.3.11 Power interface. Direct current power shall be applied to the distribution unit by any one of the connected radio sets when the radio is turned on, via pin B of J501, J503, J509, or J511. An auxiliary power input, J508, shall also be provided to permit use of the intercom without radios. All power inputs shall be diode isolated to prevent cross coupling between sources. After a radio is turned on, power-up of the intercom is achieved by setting the power circuit breaker on the distribution unit to the ON position (see 4.6.3).

3.3.3.12 Power distribution. The distribution unit shall accept supply voltage from pin B of J501, J503, J509, J511 or J508. The source shall be diode isolated to prevent cross coupling of power between connectors (see 4.6.3).

3.3.3.13 Retransmission switch. When the retransmission switch is placed in the RETRANSMISSION position, pin T of J501 shall be connected to pin S of J503, and pin S of J501 shall be connected to pin T of J503. Likewise, receiver audio from pin K of J501 shall be directed to pin U of J503, and receiver audio from pin K of J503 shall be directed to pin U of J501. Receiver audio shall be available for monitoring by crew members at any crew control unit by selecting the appropriate monitor switches; however it shall not be possible for any crew control unit to initiate a radio transmission on the radios connected to J501 or J503 when in the RETRANSMISSION mode (see 4.6.3).

3.3.4 Parts. The selection of parts shall be based on the requirements of MIL-P-11268, MIL-E-4158 and MIL-STD-454. Parts selection shall be in accordance with the following order of precedence (see 4.6.1):

a. Standard parts as defined by MIL-E-4158.
b. The TACOM Military Component Supply Directory.
c. Ordnance.
d. Federal Specifications and Standards - as defined by Appendix II-A of DOD 4120.3-M.
e. Industry Standards.
f. Manufacturer's catalogue items.

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3.3.4.1 Electrical connectors. Connectors shall be in accordance with MIL-STD-454, requirement 10. The soldering of connectors shall be in accordance with MIL-STD-454, requirement 5. When brazing is used, it shall be in accordance with MIL-B-7883 (see 4.6.1).

3.3.5 Processes.

3.3.5.1 Heat treatment. Heat treatment of aluminum alloys shall be in accordance with MIL-H-6088. Heat treatment of steel alloys shall be in accordance with MIL-H-6875 (see 4.6.1).

3.3.5.2 Welding repairs. Welding repairs of any type or class shall be made only when, and to the extent, specifically authorized by the applicable drawings, specifications, or welding procedures, which have been approved by the Government (see 4.6.1).

3.4 Performance.

3.4.1 Supply range. The distribution unit shall operate within specifications over a supply voltage range of 20 to 30 volts (V) dc. The dc power requirement shall not exceed 20 watts at 26 V dc. The distribution unit shall not be damaged by operation from a power supply operating at the worst case limits of MIL-STD-1275 (see 4.6.4.1).

3.4.2 Radio receive audio input impedance. The input impedance between pins A and K of J501, J503, J509 and J511 shall be 150 ohms $\pm$ 15 percent (%) each when measured at 1000 Hertz (Hz) (see 4.6.4.2).

3.4.3 Radio receive gain. The gain from pins A and K of J501 to pin J of J504, J505, J506 and J507 shall be 417 $\pm$ 23 millivolts (mV) per V when measured at 1000 Hz. The gain from pins A and K of J503 to pin M of J504, J505, J506 and J507; the gain from pins A and K of J509 to pin T of J504, J505, J506 and J507; and the gain from pins A and K of J511 to pin S of J504, J505, J506 and J507 shall also be 417 $\pm$ 23 mV per V when measured at 1000 Hz (see 4.6.4.3).

3.4.4 Radio receive frequency response. With a 0.22 V root mean square (rms) signal applied at the frequencies listed below, the output voltage shall be as specified below (see 4.6.4.4):

Frequency (Hz)	Output level (mV)
150	8 $\pm$ 3
300	80.5 $\pm$ 20
1000	91.7 $\pm$ 5
3500	97.9 $\pm$ 2.2
5000	98.3 $\pm$ 2.2

3.4.5 Radio receive distortion. The radio receive buffer shall perform as specified with a 0.22 V rms input at pin K of J501, J503, J509 and J511 with less than 2% total harmonic distortion (thd) when measured at 1000 Hz (see 4.6.4.5).

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3.4.6 Radio receive crosstalk. The output from any non-driven radio buffer shall be at least 50 decibels (db) below the output of a driven radio buffer channel (see 4.6.4.6).

3.4.7 Radio receive buffer output impedance. The output impedance between pins J, M, T and S and pin A of J504, J505, J506 and J507 shall be 46 ohms ± 10% each when measured at 1000 Hz (see 4.6.4.7).

3.4.8 Intercom accent control. The intercom accent control shall reduce the output of the radio receive buffers 20 ± 3 db when activated (see 4.6.4.8).

3.4.9 Microphone buffer audio input impedance. The input impedance between pins K, N, U and V and pin A of J504, J505, J506 and J507 shall be 68 ohms ± 10% each, when measured at 1000 Hz (see 4.6.4.9).

3.4.10 Microphone buffer gain. The voltage gain at 1000 Hz from pin K of J504, J505, J506 and J507 to pin U of J501; pin N of J504, J505, J506 and J507 to pin U of J503; pin U of J504, J505, J506 and J507 to pin U of J509; and pin V of J504, J505, J506 and J507 to pin U of J511 shall be 11.86 ± 0.8 V per V (see 4.6.4.10).

3.4.11 Microphone buffer frequency response. The microphone buffer frequency response shall be flat ± 2 db from 300 to 3500 Hz (see 4.6.4.11).

3.4.12 Microphone buffer distortion. The microphone buffer shall perform as specified with a 35 mV, 1000 Hz signal applied at the input. The output shall contain less than 2% thd (see 4.6.4.12).

3.4.13 Microphone buffer crosstalk. The output from any non-driven microphone buffer shall be at least 50 db below the output of a driven microphone buffer (see 4.6.4.13).

3.4.14 Microphone buffer output impedance. The output impedance between pins U and A of J501, J503, J509 and J511 shall be 150 ohms ± 10% each when measured at 1000 Hz (see 4.6.4.14).

3.4.15 Retransmit gain. With the retransmit switch on RETRANS, the voltage gain from pin K of J501 to pin U of J503 and from pin K of J503 to pin U of J501 shall be 2.01 ± 0.5 V per V when measured at 1000 Hz (see 4.6.4.15).

3.4.16 Line amplifier input impedance. The input impedance at the line terminals when measured at 1000 Hz shall be 600 ohms ± 10% balanced, when in the LISTEN mode (see 4.6.4.16).

3.4.17 Line amplifier gain (listen mode). The voltage gain from the line terminals to pin R of J504, J505, J506 and J507 shall be 66.83 ± 3.06 mV per V each when measured at 1000 Hz (see 4.6.4.17).

3.4.18 Line amplifier distortion (listen mode). The line amplifier shall perform as specified with a 0.53 V rms signal applied to the input. The output shall contain less than 5% thd at 1000 Hz (see 4.6.4.18).

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3.4.19 Line amplifier gain (talk mode). The voltage gain from pin R of J504, J505, J506 and J507 to the line terminals shall be 14.8 ± 1.96 V per V each when measured at 1000 Hz with the line terminals connected to a 600 ohm load (see 4.6.4.19).

3.4.20 Line amplifier distortion (talk mode). The line amplifier shall perform as specified with a 35 mV rms, 1000 Hz signal applied to the input. The output shall contain less than 10% thd (see 4.6.4.20).

3.4.21 Abnormal load (line amplifier). The line amplifier shall not be damaged by the application of 50 V rms, 20 Hz ringing signals while in the TALK mode or the LISTEN mode (see 4.6.4.21).

3.4.22 Auxiliary amplifier input impedance. The input impedance of the auxiliary terminals shall be 600 ohms ± 15% when measured at 1000 Hz (see 4.6.4.22).

3.4.23 Auxiliary amplifier gain. The voltage gain from the auxiliary input terminals to pin R of J504, J505, J506 and J507 shall be 4.79 ± 0.183 mV per V each when measured at 1000 Hz (see 4.6.4.23).

3.4.24 Auxiliary amplifier distortion. The auxiliary amplifier shall perform as specified with a 7 V rms, 1000 Hz signal applied to the input. The output shall contain less than 5% thd (see 4.6.4.24).

3.4.25 Abnormal input (auxiliary). The auxiliary amplifier shall not be damaged by the application of a 50 V rms, 20 Hz ringing signal (see 4.6.4.25).

3.4.26 Keying. Each distribution unit shall be capable of passing a keying signal (0.5 amp inductive load) to ground from pins D, E, F and H of J504 to pin S of J501, J503, J509 and J511, respectively. With pins D, E, F and H of J504 open circuited, the resistance from pin S to pin A on J501 shall exceed 10 000 ohms. With pins D, E, F and H of J504 grounded, the resistance betweein pins S and A on J501 (radio 1 selected) shall be less than 2 ohms. The keying circuit line shall be capable of withstanding an open circuit voltage equivalent to the power supply voltage specified in 3.4.1 (see 4.6.4.26).

3.4.27 Environmental conditions. The distribution unit shall operate as specified in the environmental conditions specified herein. The distribution unit shall meet the performance requirements of this specification during exposure to designated environments while operating. The unit shall not be damaged nor its performance degraded after exposure to those environments designated nonoperating.

3.4.27.1 Temperature. The distribution unit shall operate in ambient air temperatures of -25 to 160 degrees Fahrenheit (°F) [-32 to 71 degrees Celsius (°C)]. The distribution unit shall withstand storage temperatures of -40 to 160°F (-40 to 71°C) for a duration of 6 hours (see 4.6.4.27.1).

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3.4.27.2 Low pressure (altitude). The distribution unit shall operate at altitudes to 10 000 feet (ft) [3000 meters (m)]. Nonoperating, the distribution unit shall withstand exposure to altitudes to 40 000 ft (12 000 m) (see 4.6.4.27.2).

3.4.27.3 Humidity. The distribution unit shall operate in humidity conditions of 5 to 95% relative humidity (rh) (see 4.6.4.27.3).

3.4.27.4 Fungus. The distribution unit shall be operational after exposure to the fungi specified in MIL-STD-810, method 508.3, for 90 days at a rh of 95% and a temperature of 84°F (29°C) (see 4.6.4.27.4).

3.4.27.5 Salt fog. The distribution unit shall operate when exposed to salt fog conditions as defined in MIL-STD-810, method 509.2, procedure I except the salt fog fallout shall be 25 lbs/acre/year (see 4.6.4.27.5).

3.4.27.6 Water spray. The distribution unit shall be spray tight as defined in MIL-STD-108, table II, so that when the vehicle is cleaned by hosing, the performance of the distribution unit is not degraded (see 4.6.4.27.6).

3.4.27.7 Sand and dust. The distribution unit shall not be degraded by exposure to blowing sand or dust (see 4.6.4.27.7).

3.4.27.8 Eplosive atmosphere. The distribution unit shall not cause ignition of an ambient explosive gaseous mixture of air when operating in such an atmosphere (see 4.6.4.27.8).

3.4.27.9 Vibration. The distribution unit shall pass the vibration requirement of method 514.3, category 8, procedure I of MIL-STD-810, with the test level as specified in figure 3 (see 4.6.4.27.9).

3.4.27.10 Shock, ballistic. The distribution unit shall withstand a shock test for light weight equipment in accordance with MIL-S-901 (see 4.6.4.27.10).

3.4.27.11 Electromagnetic interference. When specified (see 6.2), electromagnetic interference (EMI) suppression shall be in accordance with MIL-STD-461 for class A3 vehicles and equipment (see 4.6.4.27.11).

3.5 Painting. Unless otherwise specified herein, surface treatments, coatings, and finishes for the distribution unit shall conform to MIL-E-4158. Aluminum surfaces shall be cleaned and treated per MIL-C-5541 and primed with MIL-P-53022. Painting shall be Forest Green, color No. 34079 per FED-STD-595, epoxy-polyamide chemical agent resistant coating per MIL-C-22750, type I. Apply per MIL-F-18264 (see 4.5.2 and 4.6.2).

3.6 Identification and marking. The exterior markings shall be in accordance with MIL-STD-130 and MIL-STD-454, except for safety markings, and shall conform to MIL-STD-642 (see 4.5.2 and 4.6.2).

3.6.1 Safety markings. Unless otherwise specified (see 6.3), safety plates and decals shall be in accordance with applicable drawings (see 4.5.2 and 4.6.2).

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3.6.2 Nameplate and product marking. Parts and assemblies shall be marked for identification in accordance with MIL-M-13231. Whenever practicable, parts shall be so mounted that their identification marking will be readily visible (see 4.5.2 and 4.6.2).

3.6.2.1 Nameplates. Nameplates marked in accordance with MIL-STD-130 shall be permanently attached to the equipment. Nameplates shall be in accordance with Drawing 12310238 (see 4.5.2 and 4.6.2).

3.6.2.2 Part numbers. The part number shall be assigned and governed by the requirements of DOD-STD-100 and shall be the same as, or shall include, the number of the drawing which depicts the part (see 4.5.2 and 4.6.2).

3.7 Workmanship. The distribution unit shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of MIL-P-11268 (see 4.5.2 and 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2 Parts and components. The supplier, when requested, shall make available to the Government at the time of any Government inspection, legible drawings and printed specifications to which the end item or any of its parts/components were manufactured. Such drawings and specifications shall be annotated to reflect the latest revision incorporated therein. Upon completion of inspection(s) by the Government, all drawings and specifications will be returned to the supplier. The Government reserves the right to inspect end items or any parts/components during all manufacturing processes and reject such material that does not conform to either Government or supplier drawings/specificatins. All deficiencies detected during any supplier or Government inspection (end item or in-process) shall be corrected by the supplier. During any Government inspection, the supplier shall provide inspection assistance upon request.

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4.1.3 Quality assurance requirements (QAR)/supplementary quality assurance provisions (SQAP). In the event that ordnance or MS drawings referenced in this document or lower tier drawings applicable to the referenced drawings have QAR's or SQAP's applicable (see 6.2), the supplier shall inspect the parts/components in accordance with the QAR and SQAP requirements. In addition, General Quality Assurance Requirements (STA Form 4452) also apply in conjunction to the QAR/SQAP requirements. The supplier shall submit QARs/SQAPs to subcontractors with purchase orders for items that have QAR/SQAP requirements.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).
b. Quality conformance inspections (see 4.5).
   1. Acceptance tests (see 4.5.3.1).
   2. Control tests (see 4.5.3.2).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 ± 18°F (23 ± 10°C)
b. Barometric pressure $28.5 \begin{smallmatrix} +2.0 \\ -3.0 \end{smallmatrix}$ inches Hg $(725 \begin{smallmatrix} +50 \\ -75 \end{smallmatrix}$ mm Hg)
c. Relative humidity 50 ± 30 percent

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply distribution units that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two distribution units. Preproduction inspection shall consist of inspection as specified in table I.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two from the first ten distribution units produced under the production contract for initial production. Initial production units shall be inspected as specified in table I.

4.4.2.1 In-process inspection. During fabrication of the first production distribution unit, in-process inspections will be conducted by Government representatives to evaluate conformance of materials and workmanship to requirements of specified documents. The distribution unit and all components shall be available for inspection. These inspections shall be made at the contractor or subcontractor's facilities prior to application of primer and paint. Processing and welding procedures, quality system, inspection records, calibration procedures, radiographic procedures and welder certifications will be reviewed and evaluated during the in-process inspections.

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4.4.3 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article testing is completed to the satisfaction of the Government.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance	
				Acceptance	Control
Materials and construction	3.2.1 thru 3.2.7, 3.3 and 3.3.4 thru 3.3.5.2	4.6.1	X		X
Defects (see table II)	3.3.1, 3.3.2 and 3.5 thru 3.7	4.6.2	X	X	
Size and weight	3.3.2	4.6.3	X		X
Interface definition	3.3.3.1	4.6.3	X	X	
On-off switch	3.3.3.2	4.6.3	X	X	
Line jacks	3.3.3.3	4.6.3	X	X	
Auxiliary input	3.3.3.4	4.6.3	X	X	
Intercom cable assembly	3.3.3.5	4.6.3	X	X	
Power indicator	3.3.3.6	4.6.3	X	X	
Intercom accentuated switch	3.3.3.7	4.6.3	X	X	
Radio transmission selector switch	3.3.3.8	4.6.3	X	X	
Radio retransmission switch	3.3.3.9	4.6.3	X	X	
Line switch	3.3.3.10	4.6.3	X	X	
Power interface	3.3.3.11	4.6.3	X	X	
Power distribution	3.3.3.12	4.6.3	X	X	
Retransmission switch	3.3.3.13	4.6.3	X	X	
Supply range	3.4.1	4.6.4.1	X	X	
Radio receive audio input impedance	3.4.2	4.6.4.2	X	X	
Radio receive gain	3.4.3	4.6.4.3	X	X	

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TABLE I. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance	
				Acceptance	Control
Radio receive frequency response	3.4.4	4.6.4.4	X	X	
Radio receive distortion	3.4.5	4.6.4.5	X	X	
Radio receive crosstalk	3.4.6	4.6.4.6	X	X	
Radio receive buffer output impedance	3.4.7	4.6.4.7	X	X	
Intercom accent control	3.4.8	4.6.4.8	X	X	
Microphone buffer audio input impedance	3.4.9	4.6.4.9	X	X	
Microphone buffer gain	3.4.10	4.6.4.10	X	X	
Microphone buffer frequency response	3.4.11	4.6.4.11	X	X	
Microphone buffer distortion	3.4.12	4.6.4.12	X	X	
Microphone buffer crosstalk	3.4.13	4.6.4.13	X	X	
Microphone buffer output impedance	3.4.14	4.6.4.14	X	X	
Retransmit gain	3.4.15	4.6.4.15	X	X	
Line amplifier input impedance	3.4.16	4.6.4.16	X	X	
Line amplifier gain (listen mode)	3.4.17	4.6.4.17	X	X	
Line amplifier distortion (listen mode)	3.4.18	4.6.4.18	X	X	
Line amplifier gain (talk mode)	3.4.19	4.6.4.19	X	X	
Line amplifier distortion (talk mode)	3.4.20	4.6.4.20	X	X	
Abnormal load (line amplifier)	3.4.21	4.6.4.21	X	X	
Auxiliary amplifier input impedance	3.4.22	4.6.4.22	X	X	
Auxiliary amplifier gain	3.4.23	4.6.4.23	X	X	
Auxiliary amplifier distortion	3.4.24	4.6.4.24	X	X	

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TABLE I. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance	
				Acceptance	Control
Abnormal input (auxiliary)	3.4.25	4.6.4.25	X	X	
Keying	3.4.26	4.6.4.26	X	X	
Temperature	3.4.27.1	4.6.4.27.1	X		
Low pressure (altitude)	3.4.27.2	4.6.4.27.2	X		
Humidity	3.4.27.3	4.6.4.27.3	X		
Fungus	3.4.27.4	4.6.4.27.4	X		
Salt fog	3.4.27.5	4.6.4.27.5	X		
Water spray	3.4.27.6	4.6.4.27.6	X		
Sand and dust	3.4.27.7	4.6.4.27.7	X		
Explosive atmosphere	3.4.27.8	4.6.4.27.8	X		
Vibration	3.4.27.9	4.6.4.27.9	X		
Shock, ballistic	3.4.27.10	4.6.4.27.10	X		
Electromagnetic interference	3.4.27.11	4.6.4.27.11	X		

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the distribution units of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

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TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly, incomplete (see 3.3, 3.3.3.2 through 3.3.3.5).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
103	Electrical connections, improper (see 3.3.4.1).	Visual
104	Welding repairs, improper (see 3.3.5.2).	Visual
105	Identification marking, missing, incomplete, or not legible (see 3.6 through 3.6.2.2).	Visual
106	Faulty workmanship affecting performance (see 3.7).	Visual
Minor	AQL 2.5% Defective	
201	Lubrication, improper application or level (see 3.2.4).	Visual
202	Wiring and cables, defective, missing, improper installation, coding or protection (see 3.3).	Visual
203	Fasteners, improper assembly or torque (see 3.3).	Visual and SIE 1/
204	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
205	Paint, improper application or color (see 3.5).	Visual
206	Identification marking, improper location or size, defective (see 3.6 through 3.6.2.2).	Visual
207	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests. Each distribution unit shall be subjected to the tests specified in table I.

4.5.3.2 Control tests. Control tests shall be conducted on one distribution unit from each lot of 250 units consecutively produced, except that not more than one test shall be performed in a three-month period, nor less than one test in a six-month period. The distribution unit shall be subjected to the tests specified in table I.

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4.5.3.3 Failure. Failure of any distribution unit to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2.1 thru 3.2.7, and 3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vender catalogs and certificatoins, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.3.1, 3.3.2, and 3.5 through 3.7 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.6.3 Features. Conformance to 3.3.1 thru 3.3.3.11, 3.4.1, and 3.4.2 shall be determined by exercising the distribution unit, and by qualitative observation sufficient to demonstrate that specified characteristics and features are present and functional.

4.6.4 Performance tests. Except as otherwise specified herein, equipment performance shall be measured with a 333 ohm resistor in parallel with an 825 ohm -0.27 microfarad series combination connected between pin R and signal ground in order to simulate system operation. Except as otherwise specified herein, gain measurements shall be made with equipment volume controls set to maximum.

4.6.4.1 Supply range. To determine conformance to 3.4.1, the equipment shall be operated from a 26 V dc power supply and the current shall be measured and shall not exceed 0.77 amperes. The equipment shall be operated from a supply operating at the worst case limits of MIL-STD-1275. The supply need not exhibit the time requirements of MIL-STD-1275, however, the applicable curves shown by MIL-STD-1275 shall be contained with the resulting voltage-time characteristic of the power supply used. The equipment shall not be damaged or degraded.

4.6.4.2 Radio receive audio input impedance. To determine conformance to 3.4.2, measure the input impedance between pins A and K of J501, J503, J509 and J511 at 1000 Hz.

4.6.4.3 Radio receive gain. To determine conformance to 3.4.3, apply a 0.22 V rms, 1000 Hz signal to pins A and K of J501. With the output taken at pin J of J504, J505, J506 and J507, each output shall measure 91.7 $\pm$ 5.0 mV into an open circuit. With the same signal applied to pins A and K of J503, J509 and J511 and the output taken at pin M of J504, J505, J506 and J507; pin T of J504, J505, J506 and J507; and pin S of J504, J505, J506 and J507, respectively, each output shall measure 91.7 $\pm$ 5.0 mV into an open circuit. This circuit corresponds to a gain of 416 $\pm$ 23 mV per V.

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4.6.4.4 Radio receive frequency response. To determine conformance to 3.4.4, apply a constant amplitude variable frequency signal and with the output taken as specified in 4.6.4.3, the output voltage shall be as specified in 3.4.4.

4.6.4.5 Radio receive distortion. To determine conformance to 3.4.5, apply a 0.22 V rms, 1000 Hz signal to pin K of J501. With the output taken at pin J of J504, J505, J506 and J507, the distortion shall be less than 2% at each output. With the same signal applied to pin K of J503, J509 and J511 and the output taken at pin M of J504, J505, J506 and J507; pin T of J504, J505, J506 and J507; and pin S of J504, J505, J506 and J507, respectively, the distortion shall be less than 2% at each output.

4.6.4.6 Radio receive crosstalk. Under the conditions specified in 4.6.4.5, the output from any non-driven channel shall be at least 50 db below the output of a driven channel to determine conformance to 3.4.6.

4.6.4.7 Radio receive buffer output impedance. To determine conformance to 3.4.7, measure the output impedance between pins J, M, T and S and pin A of J504, J505, J506 and J507 at 1000 Hz.

4.6.4.8 Intercom accent control. To determine conformance to 3.4.8, perform the test specified in 4.6.4.3 except the intercom accent switch shall be set to "ON". Output power shall be 20 $\pm$ 3 db below the level observed in 4.6.4.3.

4.6.4.9 Microphone buffer audio input impedance. To determine conformance to 3.4.9, measure the input impedance between pins K, N, U and V and pin A of J504, J505, J506 and J507 at 1000 Hz.

4.6.4.10 Microphone buffer gain. To determine conformance to 3.4.10, apply a 35 mV signal to pin K of J504, J505, J506 and J507. With the output taken at pin U of J501, each output shall measure 415 $\pm$ 28 mV into an open circuit at 1000 Hz. With the same signal applied to pin N of J504, J505, J506 and J507; pin U of J504, J505, J506 and J507; pin V of J504, J505, J506 and J507; and output taken at pin U of J503, J509 and J511, respectively, each output shall measure 415 $\pm$ 28 mV into an open circuit at 1000 Hz. This output corresponds to a gain of 11.86 $\pm$ 0.8 V per V.

4.6.4.11 Microphone buffer frequency response. To determine conformance to 3.4.11, apply a constant amplitude variable frequency signal and with the output taken as specified in 4.6.4.10, the output voltage shall be as specified below:

Frequency (Hz)	Output level (db)
300	$\pm$ 2
1000	0 (reference 415 $\pm$ 28 mV)
3500	$\pm$ 2

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4.6.4.12 Microphone buffer distortion. To determine conformance to 3.4.12, apply a 35 mV, 1000 Hz signal to pin K of J504, J505, J506 and J507. With the output taken at pin U of J501, the distortion shall be less than 2%. With the same signal applied to pin N of J504, J505, J506 and J507; pin U of J504, J505, J506 and J507; pin V of J504, J505, J506 and J507; and output taken at pin U of J503, J509 and J511, respectively, the distortion shall be less than 2%.

4.6.4.13 Microphone buffer crosstalk. Under the conditions specified in 4.6.4.12, the output from any non-driven channel shall be at least 50 db below the output from a driven channel to determine conformance to 3.4.13.

4.6.4.14 Microphone buffer output impedance. To determine conformance to 3.4.14, measure the output impedance between pins U and A of J501, J503, J509 and J511 at 1000 Hz.

4.6.4.15 Retransmit gain. To determine conformance to 3.4.15, apply a 0.22 V rms, 1000 Hz signal to pin K of J501 and pin K of J503 with the retransmit switch on RETRANS. With the output taken at pin U of J503 and pin U of J501 respectively, each output shall measure 442 $\pm$ 110 mV into an open circuit. This output corresponds to a gain of 2.01 $\pm$ 0.5 V per V.

4.6.4.16 Line amplifier input impedance. To determine conformance to 3.4.16, measure the input impedance at the line terminals at 1000 Hz, when in the LISTEN mode.

4.6.4.17 Line amplifier gain (listen mode). To determine conformance to 3.4.17, apply a 0.53 V rms, 1000 Hz signal to the line terminals. With the output taken at pin R of J504, J505, J506 and J507, the output voltage shall measure 35.4 $\pm$ 1.6 mV. This output corresponds to a gain of 66.83 $\pm$ 3.06 mV per V.

4.6.4.18 Line amplifier distortion (listen mode). To determine conformance to 3.4.18, apply a 0.53 V rms, 1000 Hz signal to the line terminals. With the output taken at pin R of J504, J505, J506 and J507, the distortion shall not exceed 5%.

4.6.4.19 Line amplifier gain (talk mode). To determine conformance to 3.4.19, apply a 35 mV rms, 1000 Hz signal to pin R of J504, J505, J506 and J507. With the output taken at the line terminals, the output voltage shall measure 518 $\pm$ 68.6 mV each with the output terminated into 600 ohms. This output corresponds to a gain of 14.8 $\pm$ 1.96 V per V.

4.6.4.20 Line amplifier distortion (talk mode). To determine conformance to 3.4.20, apply a 35 mV rms, 1000 Hz signal to pin R of J504, J505, J506 and J507. With the output taken at the line terminals terminated into 600 ohms, the distortion shall not exceed 10%.

4.6.4.21 Abnormal load (line amplifier). To determine conformance to 3.4.21, perform the test specified in 4.6.4.20. Apply a 50 V rms, 20 Hz signal across the line terminals. Repeat the tests specified in 4.6.4.17 through 4.6.4.20. No damage or degradation shall occur.

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4.6.4.22 Auxiliary amplifier input impedance. To determine conformance to 3.4.22, measure the input impedance of the auxiliary terminals at 1000 Hz.

4.6.4.23 Auxiliary amplifier gain. To determine conformance to 3.4.23, apply a 7 V rms, 1000 Hz signal to the auxiliary input terminals. With the output taken at pin R of J504, J505, J506 and J507, each output shall measure 33.5 ± 1.3 mV. This output corresponds to a gain of 4.79 ± 0.183 mV per V.

4.6.4.24 Auxiliary amplifier distortion. To determine conformance to 3.4.24, apply an input signal of 7 V rms, 1000 Hz to the auxiliary input terminals. With the output at pin R of J504, J505, J506 and J507, the distortion shall not exceed 5%.

4.6.4.25 Abnormal input (auxiliary). To determine conformance to 3.4.25, perform the tests specified in 4.6.4.23 and 4.6.4.24, except apply 50 V rms, 20 Hz to the auxiliary input terminals. No damage or degradation shall occur.

4.6.4.26 Keying. To determine conformance to 3.4.26, pass a keying signal (0.5 amp inductive load) to ground from pins D, E, F and H of J504 to pin S of J501, J503, J509 and J511, respectively. With pins D, E, F and H of J504 open circuited, the resistance from pin S to pin A on J501 shall exceed 10 000 ohms. With pins D, E, F and H of J504 grounded, the resistance between pins S and A on J501 (radio 1 selected) shall be less than 2 ohms. The keying circuit shall be opened and the keying line subjected to power supply voltages per 4.6.4.1 via an inductive-resistive device (relay coil or equivalent) resistance of 56 ohms ± 10% and inductance of 10 milliHenry ± 20%. The keying circuit shall be exercised and no degradation shall occur.

4.6.4.27 Environmental tests.

4.6.4.27.1 Temperature. To determine conformance to 3.4.27.1, the distribution unit shall be tested as follows:

a. Operation.

1. Low temperature. The distribution unit shall be tested in accordance with MIL-STD-810, method 502.2, procedure II at -25°F (-32°C).

2. High temperature. The distribution unit shall be tested in accordance with MIL-STD-810, method 512.2, procedure II at 160°F (71°C).

b. Storage.

1. Low temperature. The distribution unit shall be tested in accordance with MIL-STD-810, method 502.2, procedure I at -40°F (-40°C) for 6 hours.

2. High temperature. The distribution unit shall be tested in accordance with MIL-STD-810, method 501.2, procedure I at 160°F (71°C) for 6 hours.

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4.6.4.27.2 Low pressure (altitude). To determine conformance to 3.4.27.2, the distribution unit shall be tested in accordance with MIL-STD-810, method 500.2 as follows:

a. Procedure I. The air pressure shall be equivalent to an altitude of 40 000 ft (12 000 m).

b. Procedure II. The air pressure shall be equivalent to an altitude of 10 000 ft (3000 m).

4.6.4.27.3 Humidity. To determine conformance to 3.4.29.3, the distribution unit shall be tested in accordance with MIL-STD-810, method 507.2, procedure II. The conditions shall conform to figure 507.2-2, cycle 5, except the humidity limits shall be 5 and 95% and the number of test cycles shall be ten. Equipment shall not be moved from the humidity chamber for measurements or power left on after measurements have been completed. If any unit fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the unit still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the unit fails to meet applicable requirements after final conditioning and adjustment, it does not pass the test. The test shall be performed with the cover off.

4.6.4.27.4 Fungus. To determine conformance to 3.4.27.4, the distribution unit shall be subjected to 90 days of the fungus test specified in MIL-STD-810, method 508.3. The parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall be sprayed. The distribution unit shall be subjected to the test in the open condition. The distribution unit shall show no evidence of viable fungus or corrosion. In lieu of a test when approved by the Government contract authority, certification, with supporting data, may be provided attesting that the item is constructed of materials which will not support fungus growth.

4.6.4.27.5 Salt fog. To determine conformance to 3.4.27.5, the distribution unit shall be tested in accordance with MIL-STD-810, method 509.2, procedure I, except the salt fog fallout shall be 25 lbs/acre/year.

4.6.4.27.6 Waterspray. To determine conformance to 3.4.27.6, the distribution unit shall be tested in accordance with spray tight requirements of MIL-STD-108, table II, when mounted by its normal mounting means.

4.6.4.27.7 Sand and dust. To determine conformance to 3.4.27.7, the distribution unit shall be tested in accordance with MIL-STD-810, method 510.2, procedures I and II, except the temperature during test step 5 shall be limited to 145°F (63°C) and the sand particles shall be 0.00039 to 0.039 inches (0.01 to 1.00 mm) in diameter with predominant diameters between 0.0059 and 0.0118 inches (0.15 to 0.3 mm), and with winds greater than 30 knots (55.6 km per hr). The dust shall have a density of 0.00000037 pounds per cubic foot (0.000000006 grams per cubic centimeter), with particle size 0.0000039 to 0.00039 inches (0.0001 to 0.01 mm) in diameter, and winds at 35 knots (64.8 km per hr).

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4.6.4.27.8 Explosive atmosphere. To determine conformance to 3.4.27.8, the distribution unit shall be tested in accordance with MIL-STD-810, method 511.2, procedure I.

4.6.4.27.9 Vibration. To determine conformance to 3.4.27.9, the distribution unit shall be tested in accordance with MIL-STD-810, method 514.3, category 8, procedure I, with the test level as specified in figure 3. The distribution unit shall be operated during the test by injecting a 1000 Hz tone into all radio input channels. The output power shall be monitored and shall remain constant. The output tone shall be monitored and there shall be no repetitive break up or garbling. Channels shall be periodically reselected. A sinusoidal cycling time of 30 minutes per 1000 miles or a maximum of 3 hours is required. Sweep time is 15 minutes from 5 to 500 to 5 Hz.

4.6.4.27.10 Shock, ballistic. To determine conformance to 3.4.27.10, the distribution unit shall be tested on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. A distribution unit mounted on a bracket assembly conforming to Drawing SC-D-48664 shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the unit. The test shall consist of a total of 9 blows: one each 1 foot blow, 3 foot blow, and 5 foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the unit under test is permissible. Upon completion of the tests and before repairs are made, the unit shall show no degradation of performance and any physical damage shall be minor only, as classified in MIL-STD-252.

4.6.4.27.11 Electromagnetic interference. To determine conformance to 3.4.27.11, the electromagnetic interference shall be tested in accordance with the applicable provisions of MIL-STD-462.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Item definition. The distribution unit is the main junction box between the intercommunication control units and the radio system installed in the vehicle, providing two-way communication between crew members and external personnel. In addition, a retransmission switch on the distrubtion unit enables two radios to be used alternately as a radio relay.

6.1.1 Intended use. This distribution unit is intended to be used in, but not limited to, the M981 Fire Support Team Vehicle (FISTV).

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of applicable drawings.
c. If electromagnetic interference (EMI) suppression is required (see 3.4.27.11).
d. If first article samples are not required (see 3.1).
e. If safety markings shall be other than as specified (see 3.6.1).
f. If responsibility for inspection shall be other than as specified (see 4.1).
g. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).
h. If SQAP's or QAR's are required (see 4.1.3).
i. If inspection conditions shall be other than as specified (see 4.3).
j. If preproduction inspection is required (see 4.4.1).
k. If initial inspection is not required (see 4.4.2).
l. Applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.3 Supersession data. This military specification supersedes Emerson Electric Company specification, M981PD-G-62488, dated 27 August 1984.

6.4 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged.

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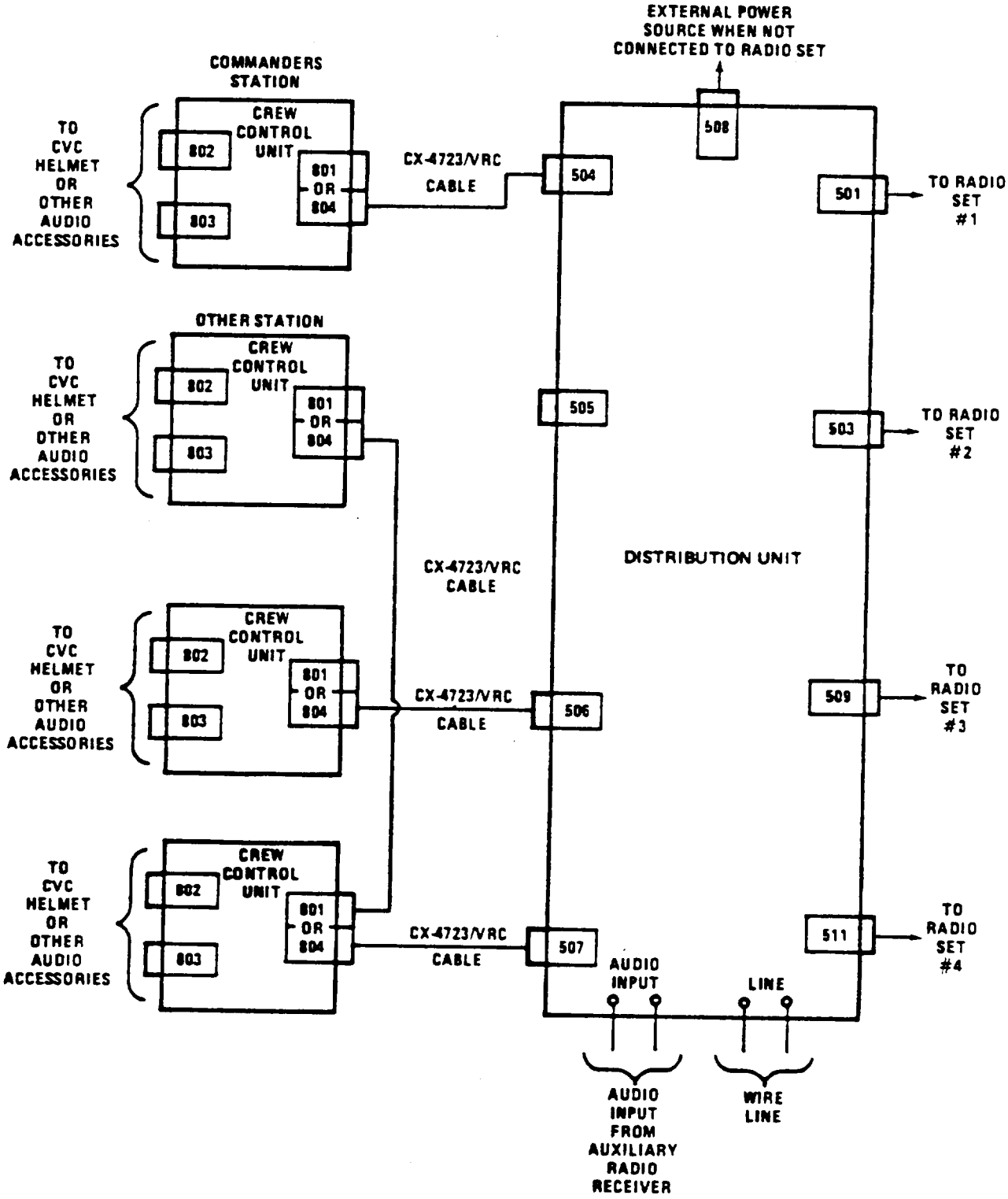


FIGURE 1. Distribution unit interface connections.

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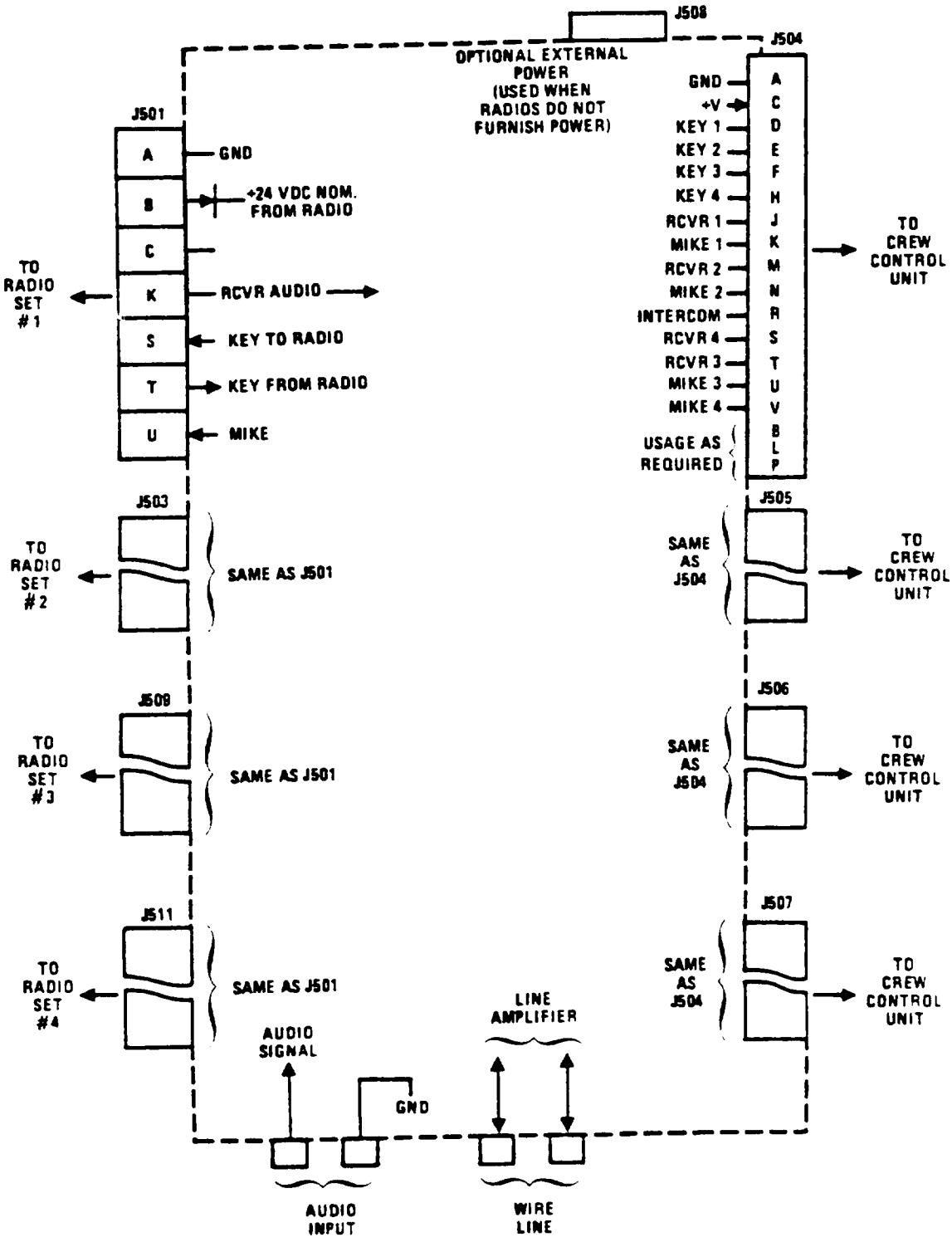
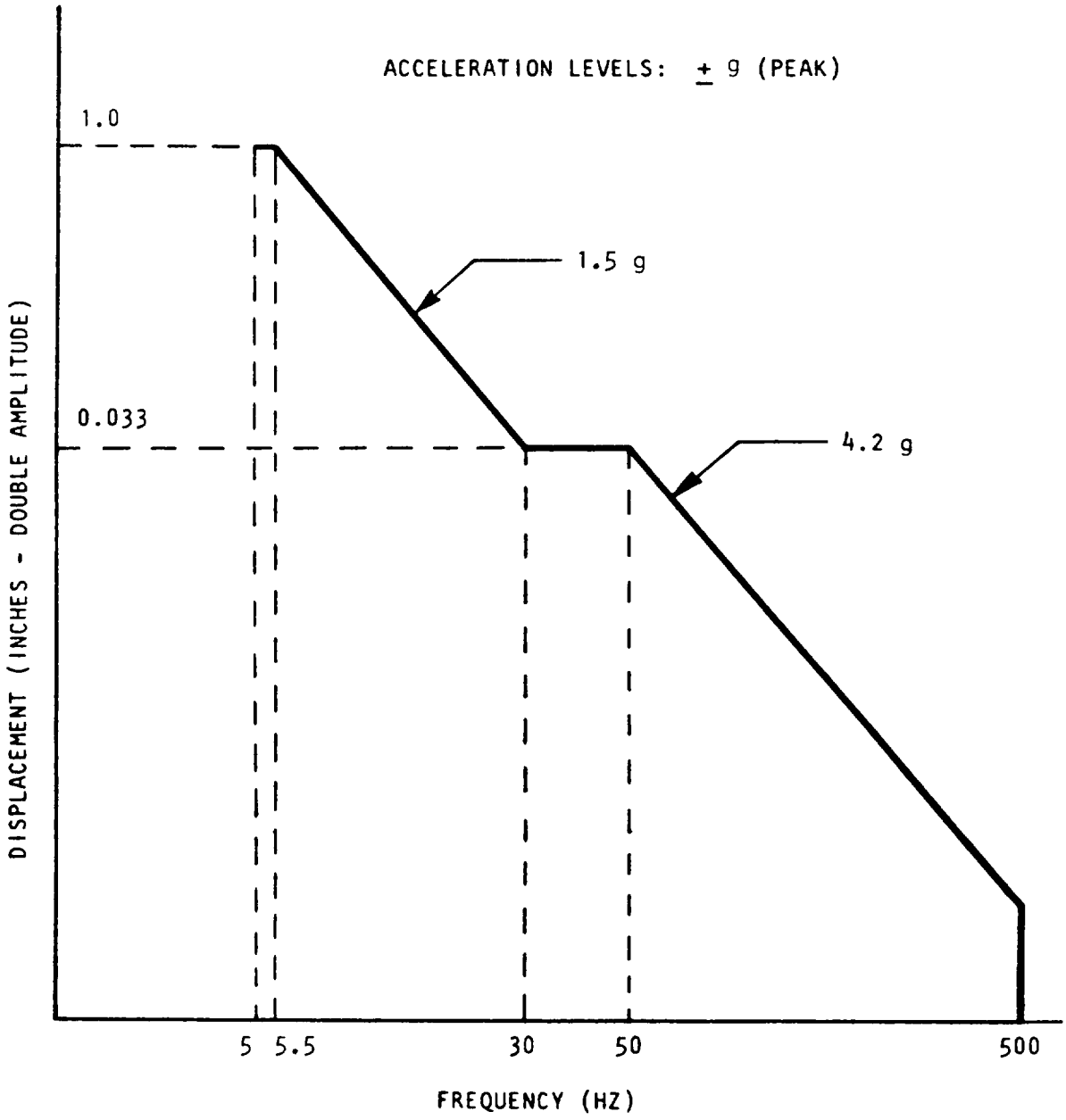


FIGURE 2. Distribution unit electrical interface.

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NOTE: All curves shall be extended to 2 Hz. when test item resonances below 5 Hz. are expected.

FIGURE 3. Vibration test curve.

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